Technical Report: Slime Mold Algorithm Bio-Inspired Simulation in Ad-Natura

SFWRENG 4G06 - Capstone Design Project

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1 Revision History

Date	Version	Notes
Mar 7, 2026	1.0	Initial report: architecture, implemented behaviors, design rationale

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2 Introduction

This report documents the design and implementation of the slime mold simulation system in *Ad-Natura*, a 2D environmental exploration game. The simulation models *Physarum polycephalum*, a single-celled organism capable of decentralized decision-making, adaptive network formation, and environmental sensing—without any central nervous system or governing structure [4, 1].

The system runs entirely on the GPU as a compute shader, driving 1024 autonomous agents that deposit chemical trails, sense their environment, and collectively produce emergent network structures. Beyond serving as a game mechanic, the simulation is designed as a site of inquiry: a counter-algorithmic system [3] that resists optimization in favor of organic unpredictability.

This report covers the technical architecture (Section 3), agent behavior (Section 4), trail dynamics including implemented features (Section 5), noise generation (Section 6), game integration (Section 7), design rationale (Section 8), technical challenges (Section 9), and future work (Section 10).

3 System Architecture

The simulation is implemented as a Unity compute shader (`Slime.compute`) driven by a C# controller (`Slime.cs`). It uses a ping-pong double-buffering scheme and two GPU kernels dispatched each `FixedUpdate` frame.

3.1 Data Structures

Each agent is represented by a minimal struct:

```
struct Agent {
    float2 position;   // world-space position
    float  angle;      // heading in radians
    float4 type;       // species color (unused in current build)
};
```

Agents are stored in a `StructuredBuffer` of 1024 elements, initialized with random positions and angles via a uint-based hash PRNG.

3.2 Ping-Pong Buffering

Two `RenderTexture` buffers (`TrailMap` and `TrailMapProcessed`) alternate roles each frame. The **Update** kernel reads from the current trail map and writes agent deposits to it. The **Postprocess** kernel reads from that same buffer, applies diffusion, decay, and pulse, then writes the result to the other buffer. On the next frame, the roles swap. This prevents read-write hazards on the GPU.

Both trail maps use a floating-point texture format (`RenderTextureFormat.RFloat` for attraction maps, `RenderTextureFormat.ARGBFloat` for trail maps). This is critical: trail

intensity values routinely exceed 1.0, creating a "reservoir effect" where dense trail networks store accumulated chemical signal. Clamping to $[0, 1]$ destroys this behavior entirely.

3.3 Kernel Overview

- **Update kernel** (`[numthreads(16,1,1)]`): One thread per agent. Handles sensing, turning, movement, boundary behavior, and trail deposition. Dispatched as $\lceil 1024/16 \rceil = 64$ thread groups.
- **Postprocess kernel** (`[numthreads(8,8,1)]`): One thread per texel. Handles 3×3 diffusion, entropic decay, and cytoplasmic pulse. Dispatched over the full trail map resolution.

4 Agent Behavior and Chemotaxis

Each agent performs a sense-turn-move cycle inspired by the Jones model of Physarum chemotaxis [1].

4.1 Sensing

The `sense()` function samples a 3×3 neighborhood at a point offset from the agent's position by `sensorOffsetDist` in a given direction. It sums both trail intensity and water attraction values, weighted by `waterWeight`:

$$S(\theta) = \sum_{i=-1}^{1} \sum_{j=-1}^{1} \Big[\text{trail}(p + d_\theta + (i, j)) + w \cdot \text{water}(p + d_\theta + (i, j)) \Big] \tag{1}$$

where p is the agent position, d_θ is the sensor offset vector at angle θ, and w is the water attraction weight.

4.2 Turning

Three sensor readings are taken: forward (S_F), left (S_L, at +`sensorAngleSpacing`), and right (S_R, at −`sensorAngleSpacing`). The turning logic follows:

1. If $S_F > S_L$ and $S_F > S_R$: continue straight.
2. If $S_F < S_L$ and $S_F < S_R$: turn randomly left or right (stochastic tie-breaking prevents deadlock).
3. If $S_L > S_R$: turn left by `turnSpeed` $\cdot \Delta t$.
4. If $S_R > S_L$: turn right by `turnSpeed` $\cdot \Delta t$.

This produces emergent path-following and network formation without any global coordination—each agent responds only to local chemical gradients.

4.3 Movement and Boundary Handling

Agent speed is modulated by local water attraction:

$$v = \texttt{moveSpeed} \cdot (1 + \texttt{waterSpeedBoost} \cdot \text{water}(p)) \tag{2}$$

At boundaries, agents do not simply reflect. Instead, they are assigned a new random heading and repositioned at the edge. An `edgeSpread` parameter adds slight random displacement along the boundary to prevent agents from clustering at corners—a subtle but important detail for maintaining organic boundary behavior.

4.4 Trail Deposition

Each agent deposits trail intensity proportional to local water attraction:

$$\text{deposit} = \texttt{trailWeight} \cdot (1 + \texttt{waterTrailBoost} \cdot \text{water}(p)) \tag{3}$$

This creates denser trail networks near water sources, visually emphasizing the organism's attraction to moisture.

5 Trail Dynamics

The Postprocess kernel transforms the raw trail map each frame through three sequential stages: diffusion, entropic decay, and cytoplasmic pulse.

5.1 Diffusion

A standard 3×3 box blur computes the mean trail value in the neighborhood of each texel. The result is linearly interpolated with the original value:

$$T'(x, y) = \text{lerp}\Big(T(x, y),\ \frac{1}{9} \sum_{i,j \in \{-1,0,1\}} T(x + i,\ y + j),\ \texttt{diffuseSpeed} \cdot \Delta t\Big) \tag{4}$$

This simulates the physical diffusion of chemical signal through the environment, smoothing the trail network over time.

5.2 Entropic Decay

Rather than applying a uniform evaporation rate, the system modulates decay with a two-octave noise field that varies in both space and time:

$$\text{noise}(x, y, t) = \texttt{organicNoise}\Big(\frac{x \cdot s_e}{W},\ \frac{y \cdot s_e}{H},\ t \cdot v_e\Big) \tag{5}$$

$$e(x, y, t) = \texttt{evaporateSpeed} \cdot \Delta t \cdot \big(1 + \texttt{entropyStrength} \cdot (2 \cdot \text{noise} - 1)\big) \tag{6}$$

$$T''(x, y) = T'(x, y) - e(x, y, t) \tag{7}$$

where s_e is the entropy scale, v_e is the entropy speed, W and H are texture dimensions. The noise field creates regions of faster and slower decay, producing organic, non-uniform trail dissolution. Some areas persist longer while others erode quickly, breaking the visual monotony of uniform evaporation.

This feature is grounded in the observation that real biological decay is never uniform—it is shaped by local moisture, temperature, microbial activity, and countless other environmental variables [3].

5.3 Cytoplasmic Streaming (Pulse)

Real *Physarum* exhibits rhythmic cytoplasmic streaming—the shuttle flow of protoplasm that oscillates through the organism's network, driving nutrient transport and enabling distributed decision-making [1].

The simulation reproduces this as a density-thresholded brightness oscillation:

$$d_n = \text{smoothstep}(0.3,\ 0.6,\ T''(x,y)/T_{\max}) \tag{8}$$

$$f(x,y) = \text{lerp}\big(\texttt{pulseMinFreq},\ \texttt{pulseMaxFreq},\ \text{water}(x,y)\big) \tag{9}$$

$$\phi(x,y) = (x+y) \cdot \texttt{pulseWaveScale} \tag{10}$$

$$\text{pulse}(x,y,t) = d_n \cdot \texttt{pulseAmplitude} \cdot \sin(2\pi f t + \phi) \tag{11}$$

The final trail output is:

$$T_{\text{out}}(x,y) = \max\big(T''(x,y) + \text{pulse}(x,y,t),\ 0\big) \tag{12}$$

Key design decisions:

- **Density threshold**: The smoothstep activation ensures only established trail networks pulse, not sparse background noise. This mirrors real Physarum, where streaming occurs in mature veins.
- **Water-driven frequency**: Pulse frequency increases near water sources, simulating heightened metabolic activity near nutrients.
- **Spatial phase offset**: The $(x+y)$ phase term creates the illusion of wave propagation across the network, rather than uniform blinking.
- **No clamping**: The output uses $\max(\cdot, 0)$ rather than $\text{clamp}(0,1)$, preserving the reservoir effect of accumulated chemical signal.

6 GPU Noise Implementation

GPU-based noise requires careful implementation. Standard approaches using sin()-based hash functions suffer from precision artifacts on many GPU architectures. The system uses an integer hash approach:

```
float hashNoise(float2 p) {
    uint2 i = uint2(abs(p));
    uint  h = i.x * 1597334677u ^ i.y * 3812015801u;
    h = h * h;
    return float(h) / 4294967295.0;
}
```

This provides well-distributed pseudorandom values without relying on trigonometric functions. A `valueNoise()` function interpolates between hash corners using smoothstep, and `organicNoise()` layers two octaves with halved amplitude:

$$\text{organicNoise}(x, y, t) = \text{valueNoise}(x, y, t) + 0.5 \cdot \text{valueNoise}(2x, 2y, 2t) \tag{13}$$

The result is normalized to $[0, 1]$ by dividing by 1.5. This two-octave structure adds fine detail to the noise field without the computational cost of full fractal Brownian motion, which is unnecessary for the visual effect.

7 Game Integration

While the simulation is designed as a self-contained system, it integrates with the broader game through several mechanics:

- **Water Attraction**: A water attraction map (`WaterAttractionMap`) is generated by the game's water simulation system and passed to the compute shader. Agents preferentially move toward water, deposit stronger trails near it, and pulse at higher frequencies in its proximity.
- **Decomposition**: The slime mold participates in the game's decomposition system, breaking down organic obstacles in the environment. Trail density in an area contributes to the decomposition rate of nearby objects.
- **Planned—Light Avoidance**: Real *Physarum* is negatively phototactic. A future implementation will add light sources that repel agents, creating gameplay opportunities around guiding the organism through shadowed paths.
- **Planned—Wall Climbing**: Extending the slime mold's navigable space to vertical surfaces, allowing players to use the organism to traverse otherwise inaccessible terrain.

The `SlimeMoldManager` class coordinates between the simulation and other game systems, providing a public API (`SetAttractionMaps`, `GetTrailTexture`, `GetWorldBounds`) for other scripts to interact with the slime mold.

8 Design Rationale

The design of this system is informed by both biological research and critical theory. Several deliberate choices distinguish it from conventional algorithmic implementations.

8.1 Self-Organization Without Central Control

Evelyn Fox Keller's foundational work on slime mold aggregation [4] demonstrated that scientists historically assumed *Dictyostelium* required a "pacemaker" cell to coordinate collective behavior. Keller showed that aggregation is fully self-organizing: no leader, no central governor.

This principle is encoded in the simulation's architecture. All 1024 agents follow identical local rules—sense, turn, move, deposit. Network formation, trail reinforcement, and path optimization emerge entirely from these local interactions. There is no global pathfinding, no master controller, no optimization objective.

8.2 Counter-Algorithmic Design

Bunton [3] critiques the Slime Mold Algorithm (SMA) for reducing *Physarum*'s complex, "queer" biology to a metaheuristic optimization tool—stripping away precisely the behaviors that make the organism interesting. The concept of "jamming"—blocking, improvising, knotting—is proposed as an alternative relation to algorithmic systems.

The entropic decay system (Section 5.2) functions as a form of jamming. Rather than optimizing trail networks toward efficiency, the noise-modulated decay introduces unpredictable dissolution. Trails that "should" persist may erode; regions that "should" clear may linger. The system resists convergence toward optimal solutions.

8.3 Nature's Performativity

Barad's theory of agential realism [2] argues that nature is not passive matter awaiting measurement, but actively performs—entities co-constitute through "intra-action" rather than interaction. The slime mold simulation embodies this: the organism and its environment are not separate systems that interact, but a single entangled process. Trail deposition shapes agent movement, which shapes trail deposition. The water map shapes the organism, and in the game's decomposition system, the organism reshapes its environment.

8.4 Non-Human Play

Stone [7] extends the concept of "player" beyond the human, drawing on Bennett's distributive agency to argue that nonhuman entities—bacteria, plants, algorithms—can be understood as players in an assemblage. In *Ad-Natura*, the slime mold is not an object the player controls but a co-player with its own agency: it responds to the environment, makes decisions (through chemotaxis), and produces outcomes the human player cannot fully predict or control.

This framing connects to Ruberg's concept of "after agency" [5]—game systems where the human player's control is deliberately limited. The player can influence the slime mold's environment (e.g., directing water flow), but cannot command the organism directly.

9 Technical Challenges and Lessons

9.1 The Reservoir Effect

The most consequential technical discovery during development was that trail values in floating-point `RenderTexture` buffers routinely exceed 1.0. When multiple agents deposit trails in the same region over several frames, the accumulated value grows well above the $[0, 1]$ range. This "reservoir effect" is essential to the simulation's visual quality: dense networks appear brighter and more saturated, while sparse trails fade naturally.

An early implementation clamped the output to $[0, 1]$, which flattened all dense networks to a uniform maximum brightness and eliminated the visual distinction between mature veins and new trails. The fix was simple but non-obvious: output $\max(\text{value}, 0)$ with no upper clamp.

9.2 GPU Noise Precision

Initial noise implementations using sin()-based hash functions produced visible banding artifacts on certain GPU architectures. The integer hash approach (Section 6) resolved this entirely, at no performance cost.

9.3 RenderTexture Format

The default water attraction map must use `RenderTextureFormat.RFloat`, not `ARGB32`. Using an integer format caused the attraction values to be silently quantized, producing step-like artifacts in agent movement near water boundaries.

9.4 Inspector Serialization

Unity's Inspector serializes field values independently of code defaults. When default parameter values were changed in the C# script, existing scene references retained their old serialized values. This caused confusion during development, as code changes appeared to have no effect until the component was reset manually.

10 Future Work

Three additional features are planned to deepen the simulation's biological fidelity and conceptual range:

10.1 Behavioral Drift

Currently, all agents share identical parameters (sensor distance, turn speed, move speed). Behavioral Drift will introduce per-agent variation: each agent's parameters will be initialized with slight offsets from the base values, and these offsets will evolve slowly over time via noise-driven drift.

This produces a population of individuals within a collective—some agents more exploratory, others more conservative; some faster, others slower. The emergent effect is a simulation that never fully converges to a single stable pattern, mirroring the paraconsistent logic of biological individuality described by Sinclair [6]: the organism is simultaneously individual *and* collective.

10.2 Microphone as Interface

Audio input from the player's microphone will modulate simulation parameters in real time. Amplitude may drive trail deposition strength; frequency spectrum may influence agent speed or turning behavior. This creates what Barad [2] would call an intra-action between the player's body and the simulated organism—not a control interface but a zone of mutual influence.

10.3 Consent and Resistance

A state system where the slime mold can "refuse" player input when overstimulated. If environmental changes are too rapid or too intense, the organism enters a refractory state—withdrawing, slowing, reducing responsiveness. This encodes the organism's autonomy: it is not an infinitely compliant tool but an entity with thresholds and boundaries.

11 Conclusion

The slime mold simulation in *Ad-Natura* is a GPU-accelerated, agent-based model of *Physarum polycephalum* that prioritizes biological plausibility and emergent complexity over algorithmic efficiency. Through entropic decay, cytoplasmic pulse, and careful architectural decisions, the system produces organic, unpredictable behavior from simple local rules.

The design is grounded in critical engagement with both biological research and theoretical frameworks that challenge conventional approaches to computational modeling of living systems. Rather than reducing *Physarum* to an optimization heuristic, the system attempts to honor the organism's capacity for self-organization, distributed agency, and resistance to centralized control.

References

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